Ex.no: 5b	DATA VISUALIZATION USING MATPLOTLIB (RESTAURANT SALES DATA ANALYSIS)
Date:	

AIM:

To perform Exploratory Data Analysis (EDA) and generate various plots (Line Chart, Bar Chart, Pie Chart, Histogram, Scatter Plot, Box Plot, and Multiple Line Charts) using Python's Matplotlib library for the Sales Data Analysis dataset.

ALGORITHM:

1. Start the Python environment and import necessary libraries (`pandas` for data manipulation, `matplotlib.pyplot` for visualization).
2. Load the dataset `9. Sales-Data-Analysis.csv` using `pd.read_csv()`.
3. Calculate Total Sales (Revenue) by multiplying the `Price` and `Quantity` columns.
4. Convert the `Date` column to standard datetime format and sort the dataframe.
5. Plot a **Line Chart** to display the daily total sales trend over time.
6. Plot a **Bar Chart** to represent total sales revenue aggregated by product category.
7. Plot a **Pie Chart** to visualize the proportion of sales across different payment methods.
8. Plot a **Histogram** to examine the distribution and frequency of sales transaction values.
9. Plot a **Scatter Plot** to analyze the relationship between Unit Price and Quantity Sold.
10. Plot a **Box Plot** to compare sales value distributions across different purchase types (In-store, Drive-thru, Online).
11. Plot a **Multiple Line Chart** to track sales trends for each purchase type simultaneously over time.
12. Display and save the generated plots.

SOURCE CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Load dataset and calculate Total Sales
df = pd.read_csv('9. Sales-Data-Analysis.csv')
df['Sales'] = df['Price'] * df['Quantity']
df['Date'] = pd.to_datetime(df['Date'], format='%d-%m-%Y')
df = df.sort_values('Date')

# 1. Line Chart - Daily Sales Trend
daily_sales = df.groupby('Date')['Sales'].sum()
plt.figure(figsize=(6, 4))
plt.plot(daily_sales.index, daily_sales.values, marker='o', color='#1f77b4',
linewidth=2)
plt.title("Daily Sales Trend")
plt.xlabel("Date")
plt.ylabel("Sales (in USD)")
plt.xticks(rotation=30)
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()

# 2. Bar Chart - Total Sales by Product
product_sales =
df.groupby('Product')['Sales'].sum().sort_values(ascending=False)
plt.figure(figsize=(6, 4))
plt.bar(product_sales.index, product_sales.values, color='#1f77b4', width=0.5)
plt.title("Total Sales by Product")
plt.xlabel("Product")
plt.ylabel("Sales (in USD)")
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.show()
```

```
# 3. Pie Chart - Sales Share by Payment Method
payment_sales = df.groupby('Payment Method')['Sales'].sum()
plt.figure(figsize=(6, 4))
plt.pie(payment_sales.values, labels=payment_sales.index, autopct='%1.1f%%',
startangle=140)
plt.title("Sales Distribution by Payment Method")
plt.tight_layout()
plt.show()

# 4. Histogram - Sales Value Distribution
plt.figure(figsize=(6, 4))
plt.hist(df['Sales'], bins=15, color='#1f77b4', edgecolor='black', alpha=0.7)
plt.title("Sales Value Distribution")
plt.xlabel("Sales Amount")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

# 5. Scatter Plot - Price vs Quantity
plt.figure(figsize=(6, 4))
plt.scatter(df['Price'], df['Quantity'], color='#1f77b4', alpha=0.7)
plt.title("Price vs Quantity Sold")
plt.xlabel("Unit Price")
plt.ylabel("Quantity Sold")
plt.tight_layout()
plt.show()

# 6. Box Plot - Sales by Purchase Type
plt.figure(figsize=(6, 4))
df.boxplot(column='Sales', by='Purchase Type', grid=False, ax=plt.gca())
plt.title("Sales Box Plot by Purchase Type")
plt.suptitle("")
plt.xlabel("Purchase Type")
plt.ylabel("Sales Amount")
```

```
plt.tight_layout()
plt.show()

# 7. Multiple Line Chart - Sales Trend by Purchase Type
pivot_sales = df.pivot_table(index='Date', columns='Purchase Type',
values='Sales', aggfunc='sum').fillna(0)
plt.figure(figsize=(6, 4))
for col in pivot_sales.columns:
    plt.plot(pivot_sales.index, pivot_sales[col], label=col.strip(), marker='o')
plt.title("Sales Trend by Purchase Type")
plt.xlabel("Date")
plt.ylabel("Sales (in USD)")
plt.xticks(rotation=30)
plt.legend()
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

OUTPUT:

Figure 1: Daily Sales Trend (Line Chart)

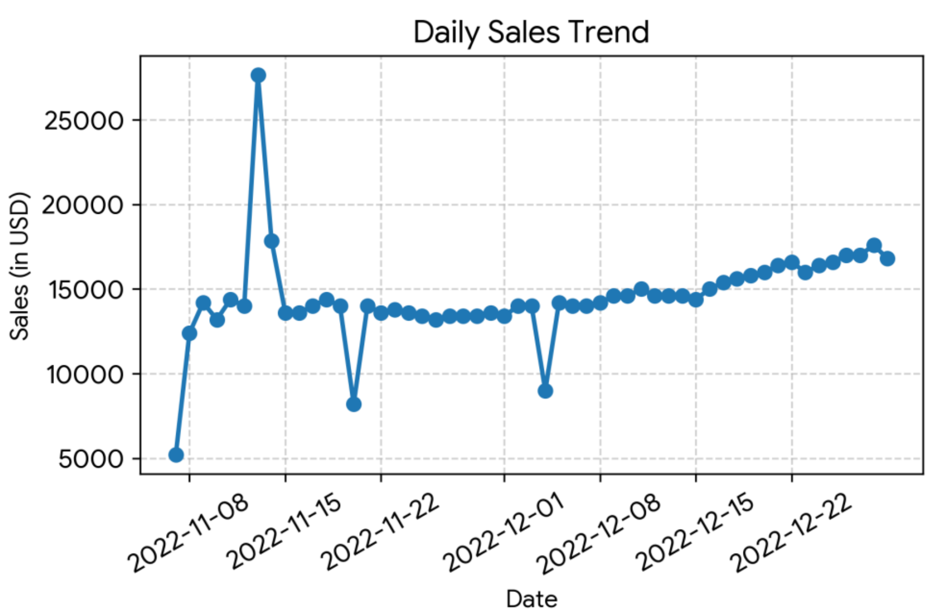


Figure 2: Total Sales by Product (Bar Chart)

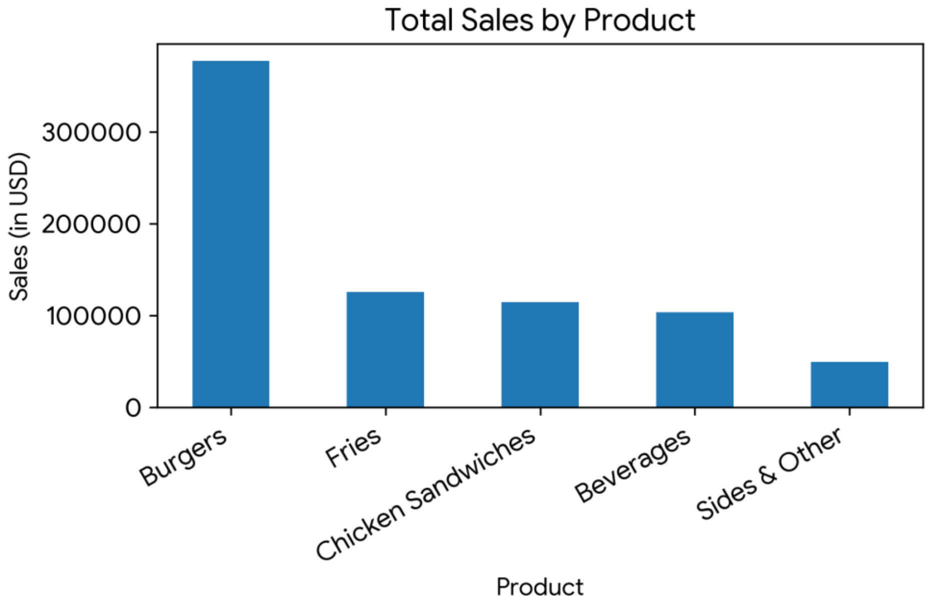


Figure 3: Sales Distribution by Payment Method (Pie Chart)

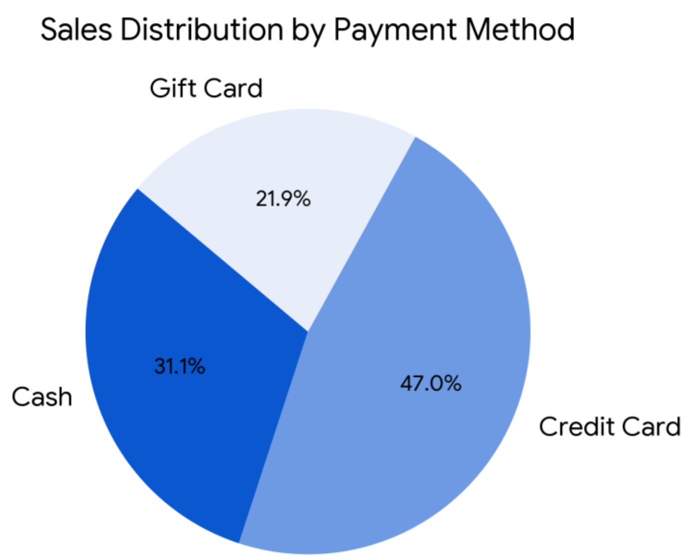


Figure 4: Sales Value Distribution (Histogram)

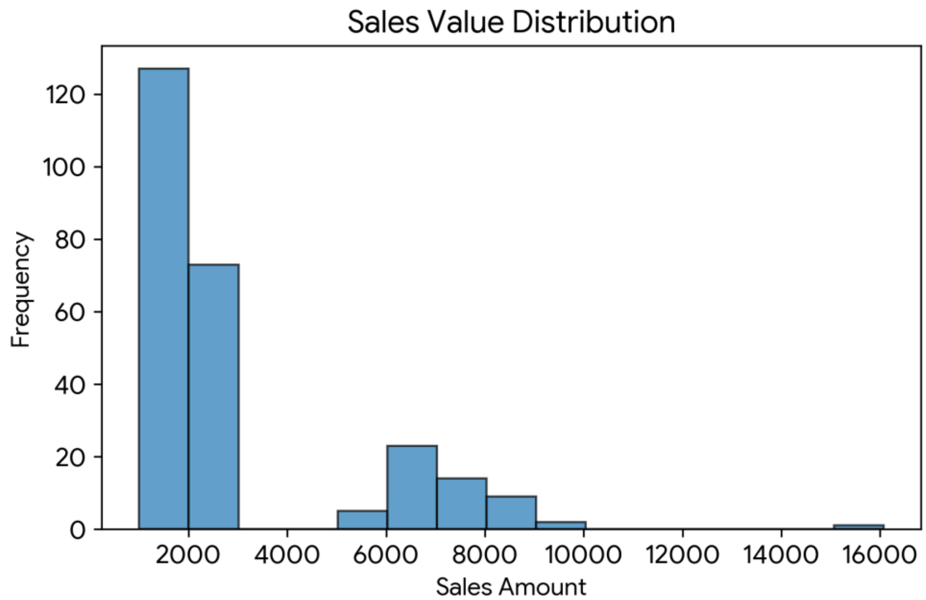


Figure 5: Price vs Quantity Sold (Scatter Plot)

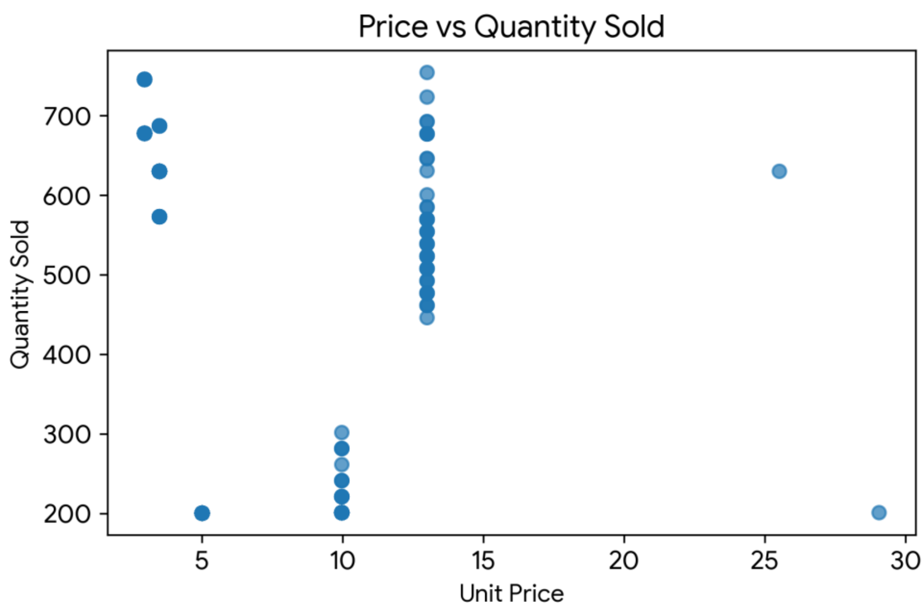


Figure 6: Sales Box Plot by Purchase Type (Box Plot)

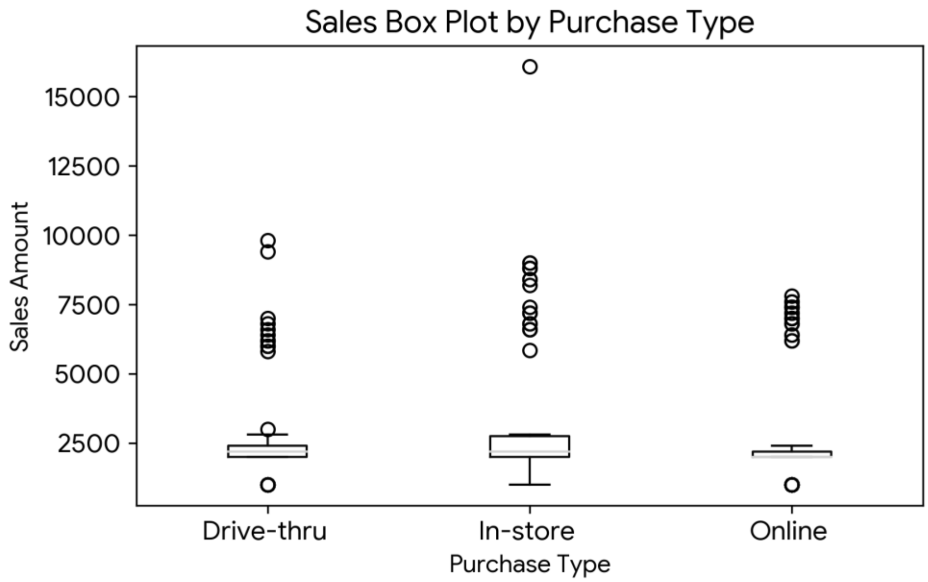
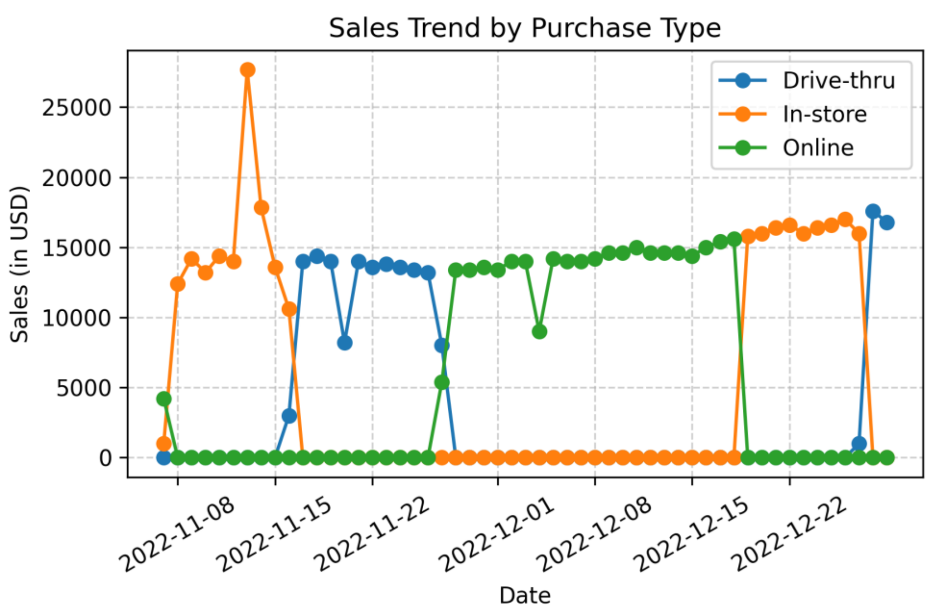


Figure 7: Sales Trend by Purchase Type (Multiple Line Chart)



RESULT: